Yahtzee!: Strategies for Maximizing Score in Solitaire Play

Group Number: 8
Name: Travis Findley
Project Topic Number: 12

Abstract

Yahtzee is a dice game played with five six-sided die and 13 turns, each turn having a maximum of three rolls. It can be played as a multi-player game or as a single player game called solitaire Yahtzee. The goal in solitaire Yahtzee is to accumulate the highest maximum score at the end of the 13 turns. The scorecard is divided into an Upper Section and Lower Section, with varying constraints, point values, and bonus points available based on roll outcomes. Assuming that a roll meets the criteria for multiple scoring boxes, the score tallying is left to the player's discretion. Should the player prioritize the Upper Section and the accompanying bonus, or should the player prioritize the Lower Section?

This analysis discovered that Upper Card Strategy was the better strategy over 5000 repetitions, producing an average total score of 214.00 with 95% confidence interval of [211.71,216.29]. These results come with some important caveats and understanding of assumptions that will be laid out through the analysis.

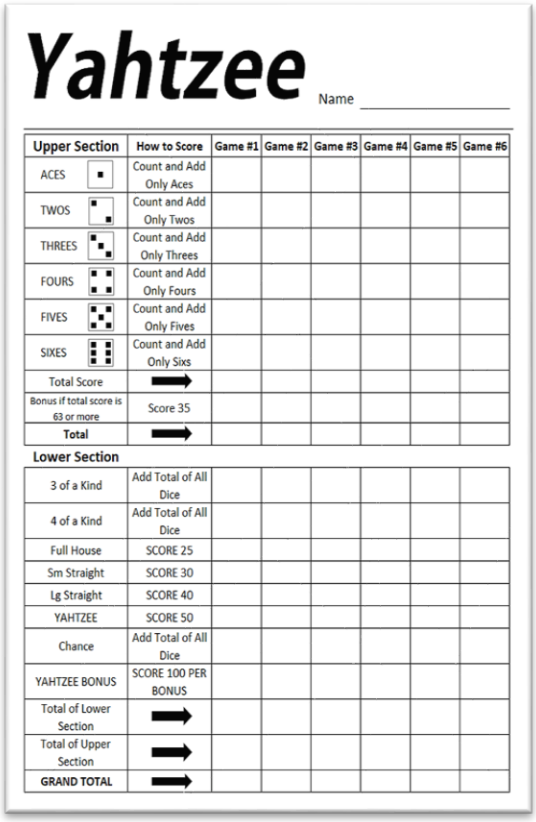
Background

Understanding Yahtzee Scoring and Relevant Rules

The Yahtzee scorecard is broken up into two distinct sections: the Upper Section and the Lower Section. The Upper Section consists of six main rows: Aces (Ones), Twos, Threes, Fours, Fives, and Sixes. For these rows, the player counts and adds only the corresponding dice accumulated during that turn (e.g. the player would record a 9 in the Threes row after concluding their three-roll turn culminating in 3, 3, 3, 1, 2). The Upper Section also contains a Bonus row, where the player is awarded 35 additional points if their total score from the six main rows is 63 or higher. The Bonus row will be one of the central focuses of this analysis.

The Lower Section consists of seven main rows and features “poker-based” constraints. The seven main rows are 3 of a Kind, 4 of a Kind, Full House, Small Straight,

Large Straight, Yahtzee, and Chance. For 3 of a Kind, 4 of a Kind, and Chance, the player adds the total of all five dice together at the end of the turn and records the score (while satisfying at least three dice the same and at least four dice the same for 3 of a Kind and 4 of a Kind). A Full House is when the player has three of one number and two of another number and is worth 25 points. A Small Straight occurs when the player has at least four sequential dice (e.g. 1, 2, 3, 4, 6) and is worth 30 points. A Large Straight occurs when the player has five sequential dice (e.g. 1, 2, 3, 4, 5) and is worth 40 points. A Yahtzee occurs when the player has all five dice the same number and is worth 50 points. The Lower Section also has a bonus row, titled Yahtzee Bonus. The player is awarded 100 points for each additional Yahtzee after filling in the Yahtzee row on the scorecard. For clarity, the scorecard rows and point values can be seen in Figure 1 below.



Yahtzee Name _____

Upper Section	How to Score	Game #1	Game #2	Game #3	Game #4	Game #5	Game #6
ACES	Count and Add Only Aces						
TWOS	Count and Add Only Twos						
THREES	Count and Add Only Threes						
FOURS	Count and Add Only Fours						
FIVES	Count and Add Only Fives						
SIXES	Count and Add Only Sixs						
Total Score	➡						
Bonus if total score is 63 or more	Score 35						
Total	➡						
Lower Section							
3 of a Kind	Add Total of All Dice						
4 of a Kind	Add Total of All Dice						
Full House	SCORE 25						
Sm Straight	SCORE 30						
Lg Straight	SCORE 40						
YAHTZEE	SCORE 50						
Chance	Add Total of All Dice						
YAHTZEE BONUS	SCORE 100 PER BONUS						
Total of Lower Section	➡						
Total of Upper Section	➡						
GRAND TOTAL	➡						

Figure 1: Standard Yahtzee scorecard

The player can also utilize each successive Yahtzee as a wild card or "Joker" via the games Joker Rules. The Joker Rules state that if both the Yahtzee row and the matching numbered row in the Upper Section have been filled, the Yahtzee may be used to fill any value in the Lower Section of the player's choosing. For example, if the player throws five 2's and the Yahtzee row and the Two's row have been filled, the player is still awarded the 100 bonus points for the additional Yahtzee and chooses Large Straight for 40 points. This

scoring format will also be reflected in the analysis. [1] With an understanding of how Yahtzee is played and scored, it is reasonable to wonder, “What is the best way to play this game?” With that question front of mind, the remainder of this paper will dive deeper into the problem, briefly look at other attempts to solve it in the past, compare two different Yahtzee strategies, and conclude with what we learned and ideas about how to improve upon the analysis in future iterations.

Main Findings

Description of the Problem

First, it is important to reiterate the distinct objective of solitaire Yahtzee and how it may differ from multi player Yahtzee. Solitaire Yahtzee is played with one player, where the objective of the player is to maximize the points in a single game. The objective in multiplayer Yahtzee is to beat your opponent(s), which may not always be in alignment with point maximization. Further, your opponent’s relative success (or failure) may impact your own decision making in a multi-player game. This analysis will only focus on solitaire Yahtzee strategy and score maximization.

With an understanding of how Yahtzee is scored and the objective of solitaire Yahtzee, we must also consider the depth of the problem. There are $6^5 = 7776$ possible dice outcomes on a single 5 dice roll turn in Yahtzee. Furthermore, there are as many as $7776^3 = 470{,}184{,}984{,}576$ possible outcomes in a three-roll turn. With such an expanse of outcomes, the dauting task is also a sandbox – there are unlimited ways to explore strategy in the game of Yahtzee! The problem is easily understood, but maybe not so easily answered: what strategy should a player use to maximize points in a game of solitaire Yahtzee? The following analysis will briefly explore past literature that pursued optimal strategies. My study will attempt to utilize the power of Monte Carlo simulation to explore two unique strategies: prioritizing the Upper Section and the accompanying Bonus or pursuing the higher point values of the Lower Section.

Literature Review

Yahtzee strategy has been the subject of several studies since its inception in 1956. Statistician Phil Woodward attempted to “solve” Yahtzee. He created a program that he called *The* Solution, targeting optimal decisions centered around expected values of dice rolls. The program had a 95% mean confidence interval of (235,356), much higher than his

personal 95% mean confidence interval of (193,252). [2] Kelly and Liese, as well as Verhoeff, looked at optimal solitaire Yahtzee strategies to maximize player. Their analysis also utilized expected values, at the category level and as total score. These separate studies both came to the same conclusion that the expected value of a solitaire Yahtzee game is 254.59. [3][4] Glenn found the same expected value of 254.59, while also exploring the average scores for other strategies. The strategies he explored include Yahtzees Only, Yahtzees and Straights, Rational Yahtzees, a greedy strategy and a couple different heuristic strategies. These strategies all performed worse than the Optimal Strategy, with a lower Expected Score and often a higher Std. Deviation. [5]

Approach

The approach in this analysis looked at two strategies: i) a strategy prioritizing the Upper Section and Bonus accumulation and ii) a strategy that pursued the higher point values of the Lower Section (Important note: the code files and algorithms were beginning to exceed the scope of the project and resources for one person. As such, both strategies attempt to make Monte Carlo runs rolls with at least one pair and higher. The strategies neglect the first roll if it has five separate numbered die. This will undoubtedly affect the analysis but must be mentioned for transparency.)

Before discussing the meaningful, albeit subtle, differences in the two strategies, it should be understood that both strategies were designed to behave logically. For example, if a Yahtzee is rolled on the first roll of a turn, the Yahtzee should be scored and the turn ended. This should be true whether the player is leaning towards prioritizing the Upper Section or the Lower Section. To understand how the strategies compare on the first roll, they can be broken down into 4 sections:

1). 5 of same dice: Both strategies accept the Yahtzee or Yahtzee Bonus and advance to next troll

2). 4 of same dice: Lower Section prioritizes 4 of a kind or 3 of a kind for high values and then moves to Upper Section. Upper Section prioritizes the Upper Section, then moves to 4 of a kind or 3 of a kind

3). 3 of same dice: Lower Section checks for Full House, then prioritizes 3 of a kind, then Upper Section. Upper Section prioritizes the Upper Section, then checks for Full House and 3 of a Kind.

4). 2 of same dice: Performs similarly to 4 and 3 of same dice, looking for point opportunities in Lower Section before checking for Upper Section. Upper Section also performs similar to previous constraints.

With the code, I leveraged the Monte Carlo simulation method to try and produce meaningful results. On the run for this project, I used a seed for reproducible results and ran 5000 iterations. Those results are laid out in the following few paragraphs.

Results

The Lower Section Strategy yielded a mean Total Score of 173.65 and a 95% confidence interval of [171.66,175.64]. In Figure 2 pictured below, it is easy to see the spikes in the average value of the mean before settling in somewhere around the 2500th repetition.

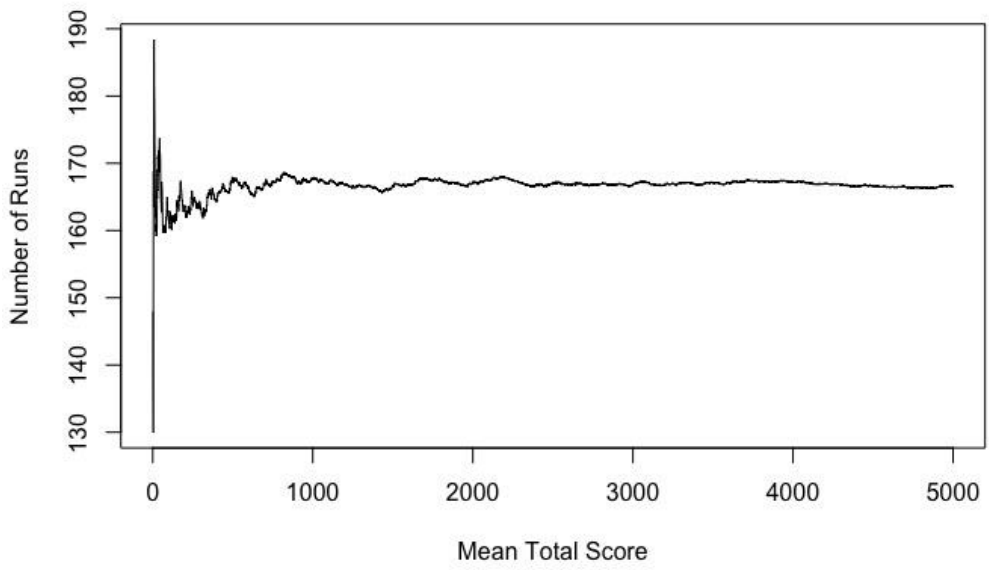


Figure 2: Mean of Total Score from the Lower Section Strategy

The Upper Section Strategy yielded a mean Total Score of 214.00 and a 95% confidence interval of [211.71,216.29]. Observations at a category level can be made with the table in Figure 3 pictured below. Most glaringly, the Upper Section strategy had a higher

mean score for the Lower Section than the Lower Section strategy, albeit with higher variance. The Upper Section strategy also strongly outperformed the Lower Section Strategy in the Upper Section. The Upper Section strategy was really buoyed by Five, Six and Bonus row, while the Lower Section strategy performed better in the 3 of a Kind, 4 of a Kind and Full House categories. The confidence intervals for both strategies can be found in their attached code file.

	Upper Section Strategy		Lower Section Strategy	
	Mean	Std Dev	Mean	Std Dev
Ones	2.9106	1.731244	2.7086	1.813269
Twos	6.0772	3.322687	5.4776	3.592225
Threes	9.4146	4.789874	8.4612	5.321744
Fours	12.5088	6.417395	11.3464	7.057845
Fives	15.8450	7.865961	8.6830	9.952403
Sixes	19.5144	9.119124	10.5720	12.022418
Bonus	22.0500	16.899841	7.1050	14.079546
Three of a Kind	16.6382	6.957398	19.4840	8.212310
Four of a Kind	11.8710	9.985044	18.6496	10.213642
Full House	15.1250	12.222490	22.7350	7.176704
Small Straight	0.4440	3.622912	0.0480	1.199160
Large Straight	4.6400	12.810278	1.6080	7.857906
Yahtzee	31.7900	24.062662	27.1700	24.908135
Chance	17.7716	4.334419	17.2360	4.179210
Yahtzee Bonus	34.2000	66.054770	23.8800	55.084957
Upper Total	88.3206	29.221605	54.3538	29.422858
Lower Total	125.6786	90.809952	119.2922	74.714829
Grand Total	213.9992	98.476470	173.6460	85.490046

Figure 3: Table for Comparison of Both Strategies at Category Level

Conclusions

My assumption beginning the project was that hard coding rules for Yahtzee was going to be quite difficult, and it proved to be much more difficult than I was able to anticipate. With five unique dice on the first roll omitted from the analysis, as well as some edge cases I inevitably missed, I am sure the analysis suffered as a result. Notably, the

tightness of the confidence intervals and the low values both the large and small straight (I believe this was greatly impacted by the aforementioned omission) were glaring holes in my undertaking of this project. The scope of this project felt much too large for one person, especially with my methodology of choice. The study did provide a fascinating look at the multiplicative power of discrete probability when multiple dice are involved. I suspect a machine learning algorithm would be better equipped to handle a simulation like this. Utilizing a neural network could possibly even expand possibilities to study the multi player game.

References

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